

5_self_admin_data

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1. Final-analysis animal IDs

```
animal_ids <- read_excel("animal_ids_corrected.xlsx",  
                        col_types = "text"  
) %>%  
  transmute(  
    subject = str_squish(subject),  
    reward  = str_to_lower(str_squish(reward)),  
    sex     = str_to_lower(str_squish(sex)),  
    day     = if_else(parse_number(day) == 1, 1L, 30L)  
  )
```

2. Find all Combined Incubation workbooks

```
incubation_files <- list.files(  
  pattern = "^Combined incubation data_.*\\.xlsx$",  
  full.names = TRUE,  
  ignore.case = TRUE  
)
```

3. Function for IVSA acquisition sheets

Infusions, active lever, inactive lever

```
read_ivsa <- function(file, sheet) {  
  
  x <- read_excel(  
    file,  
    sheet = sheet,  
    col_names = FALSE,  
    col_types = "text"  
  )  
  
  # Find the row containing "ANIMAL ID"  
  header_row <- which(  
    apply(x, 1, \(z)  
      any(  
        str_to_upper(str_squish(z)) == "ANIMAL ID",  
        na.rm = TRUE  
      )  
    )  
  )[1]  
  
  header <- x[header_row, ] %>%  
    unlist(use.names = FALSE) %>%  
    str_squish() %>%  
    str_to_upper()  
  
  # Locate animal ID and the 10 acquisition days/nights  
  id_col <- which(header == "ANIMAL ID")[1]  
  
  day_cols <- which(  
    str_detect(  
      header,  
      "^(DAY|NIGHT)\\s*(?:[1-9]|10)$"  
    )  
  )  
  
  day_nums <- parse_number(header[day_cols])  
  
  # Retain only ID + the acquisition data  
  x <- x[  
    (header_row + 1):nrow(x),  
    c(id_col, day_cols)  
  ]  
}
```

```

names(x) <- c(
  "subject",
  paste0("day_", day_nums)
)

reward <- str_extract(
  sheet,
  regex("heroin|sucrose", ignore_case = TRUE)
) %>%
  str_to_lower()

sex <- str_extract(
  sheet,
  regex("female|male", ignore_case = TRUE)
) %>%
  str_to_lower()

# Important: test "inactive" before "active"
behavior <- case_when(
  str_detect(sheet, regex("inactive", ignore_case = TRUE)) ~ "inactive",
  str_detect(sheet, regex("active", ignore_case = TRUE)) ~ "active",
  TRUE ~ "infusion"
)

x %>%
  mutate(
    subject = str_squish(subject)
  ) %>%
  pivot_longer(
    -subject,
    names_to = "day",
    names_prefix = "day_",
    values_to = "value"
  ) %>%
  mutate(
    reward = reward,
    sex = sex,
    phase = "ivsa",
    day = as.integer(day),
    behavior = behavior
  ) %>%

  # Filters to final-analysis animals without duplicating
  # animals that have both cue-reactivity days
  semi_join(
    animal_ids,
    by = c("subject", "reward", "sex")
  ) %>%

  mutate(
    value = parse_double(value)
  ) %>%

```

```
select(
  subject, reward, sex,
  phase, day, behavior, value
)
```

4. Function for cue-reactivity sheets

```
read_cue <- function(file, sheet) {

  # The first six columns contain the cohort-by-cohort raw data.
  # The additional tables to the right in some heroin sheets
  # duplicate these data, so we deliberately ignore them.
  x <- read_excel(
    file,
    sheet = sheet,
    col_names = FALSE,
    col_types = "text"
  ) %>%
    select(1:6)

  names(x) <- c(
    "subject",
    "box",
    "condition",
    "infusion",
    "active",
    "inactive"
  )

  reward <- str_extract(
    sheet,
    regex("heroin|sucrose", ignore_case = TRUE)
  ) %>%
    str_to_lower()

  sex <- str_extract(
    sheet,
    regex("female|male", ignore_case = TRUE)
  ) %>%
    str_to_lower()

  x %>%
    mutate(
```

```

# Identify Day 1 vs later incubation test
day = case_when(
  str_detect(
    subject,
    regex("^day\\s*1\\b", ignore_case = TRUE)
  ) ~ 1L,

  # Day 30, 31, 32, 33, 34 etc. are treated as Day 30
  str_detect(
    subject,
    regex("^day\\s*3\\d\\b", ignore_case = TRUE)
  ) ~ 30L,

  TRUE ~ NA_integer_
)
)%>%

# Carry the cue-test day down to the animal rows
fill(day) %>%

mutate(
  subject = str_squish(subject),
  reward = reward,
  sex = sex,
  phase = "cue_reactivity"
) %>%

# This join includes ONLY the requested cue-test day(s)
# for each final-analysis animal
semi_join(
  animal_ids,
  by = c("subject", "reward", "sex", "day")
) %>%

pivot_longer(
  c(infusion, active, inactive),
  names_to = "behavior",
  values_to = "value"
) %>%

mutate(
  value = parse_double(value)
) %>%

select(
  subject, reward, sex,
  phase, day, behavior, value
)
}

```

5. Import everything into ONE tidy dataframe

```
behavior_raw <- incubation_files %>%
  map(\(file) {

    excel_sheets(file) %>%

    map(\(sheet) {

      if (
        str_detect(
          sheet,
          regex("cue reactivity", ignore_case = TRUE)
        )
      ) {

        read_cue(file, sheet)

      } else if (
        str_detect(
          sheet,
          regex(
            "infusion|active lever|inactive lever",
            ignore_case = TRUE
          )
        )
      ) {

        read_ivsa(file, sheet)

      } else {

        NULL
      }
    }) %>%

    compact() %>%
    list_rbind()
  }) %>%

  list_rbind() %>%

  arrange(
    reward, sex, subject,
    phase, day, behavior
  )
```

```

## New names:
## New names:
## New names:
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## New names:
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## New names:
## New names:
## * ' -> '...1'
## * ' -> '...2'
## * ' -> '...3'
## * ' -> '...4'
## * ' -> '...5'
## * ' -> '...6'
## * ' -> '...7'
## * ' -> '...8'
## * ' -> '...9'
## * ' -> '...10'
## * ' -> '...11'
## * ' -> '...12'
## * ' -> '...13'
## * ' -> '...14'
## * ' -> '...15'
## * ' -> '...16'
## * ' -> '...17'
## * ' -> '...18'
## * ' -> '...19'
## * ' -> '...20'
## * ' -> '...21'

```

Run this section after the existing code has created `animal_ids` and `behavior_raw`. All files are assumed to be in `data_dir`.

```

data_dir <- "F:/My Drive/AML/rodent_incubation/analysis"

missing_ids <- read_excel(
  file.path(data_dir, "missing_animal_ids.xlsx"),
  sheet = "missing_by_row",
  col_types = "text"
) %>%
  transmute(
    subject = str_extract(str_to_upper(str_squish(subject)), "[A-Z]*\\d+"),

```

```

reward = str_to_lower(str_squish(reward)),
sex = str_to_lower(str_squish(sex)),
day = if_else(parse_number(day) == 1, 1L, 30L),
across(
  c(
    missing_infusion,
    missing_active,
    missing_inactive,
    missing_cue
  ),
  ~ str_to_lower(.x) == "true"
)
)

trap_files <- file.path(
  data_dir,
  c(
    "Heroin Incubation + D1 TRAP data 2021.xlsx",
    "Heroin Incubation + D2 TRAP data 2021.xlsx"
  )
)

clean_subject <- function(x) {
  str_extract(
    str_to_upper(str_squish(as.character(x))),
    "[A-Z]*\\d+"
  )
}

```

Read the vertically stacked infusion/active/inactive tables in one cohort tab

```

read_trap_ivsa <- function(file, sheet) {

  x <- read_excel(
    file,
    sheet = sheet,
    col_names = FALSE,
    col_types = "text"
  )

  labels <- x[[1]] %>%
    str_squish() %>%
    str_to_upper()

```

```

table_rows <- which(
  labels == "INFUSIONS" |
  str_detect(labels, "^ACTIVE\\s+LEVER") |
  str_detect(labels, "^INACTIVE\\s+LEVER")
)

map(table_rows, \(header_row) {

  behavior <- case_when(
    str_detect(labels[header_row], "^INACTIVE\\s+LEVER") ~ "inactive",
    str_detect(labels[header_row], "^ACTIVE\\s+LEVER") ~ "active",
    TRUE ~ "infusion"
  )

  first_data_row <- header_row + 2L

  # Animal rows form one contiguous block below each table header.
  block_ids <- clean_subject(x[[1]][first_data_row:nrow(x)])
  first_blank <- which(is.na(block_ids))[1]

  last_data_row <- if (is.na(first_blank)) {
    nrow(x)
  } else {
    first_data_row + first_blank - 2L
  }

  data_rows <- first_data_row:last_data_row

  header <- x[header_row, ] %>%
    unlist(use.names = FALSE) %>%
    str_squish() %>%
    str_to_upper()

  candidate_cols <- which(
    str_detect(header, "^DAY\\s*(?:[1-9]|10|OFF)$")
  )

  # Most tabs have 11 dated columns because one is a day off. In one cohort,
# the DAY OFF label is shifted by one column. Select the 10 populated
# acquisition columns and number them chronologically rather than trusting
# that label.
  populated <- map_int(
    candidate_cols,
    \(column) sum(
      !is.na(suppressWarnings(parse_double(x[[column]][data_rows])))
    )
  )

  day_cols <- candidate_cols[
    order(populated, decreasing = TRUE)[seq_len(min(10L, length(candidate_cols)))]
  ] %>%
    sort()

```

```

if (length(day_cols) != 10L) {
  stop("Could not identify 10 IVSA days in sheet: ", sheet)
}

out <- x[data_rows, c(1, 3, day_cols)]
names(out) <- c(
  "subject",
  "condition",
  paste0("day_", 1:10)
)

needed <- missing_ids %>%
  filter(.data[[paste0("missing_", behavior)]]) %>%
  distinct(subject, reward, sex)

out %>%
  mutate(
    subject = clean_subject(subject),
    condition = str_to_lower(str_squish(condition)),
    reward = "heroin"
  ) %>%
  filter(condition == "heroin") %>%
  inner_join(needed, by = c("subject", "reward")) %>%
  pivot_longer(
    starts_with("day_"),
    names_to = "day",
    names_prefix = "day_",
    values_to = "value"
  ) %>%
  transmute(
    subject,
    reward,
    sex,
    phase = "ivsa",
    day = as.integer(day),
    behavior,
    value = parse_double(value)
  )
}) %>%
  list_rbind()
}

```

Read repeated Day 1/Day 30 blocks in a consolidated cue-reactivity tab

```
read_trap_cue <- function(file, sheet = "Cue reactivity") {

  x <- read_excel(
    file,
    sheet = sheet,
    col_names = FALSE,
    col_types = "text"
  )

  row_text <- apply(
    x,
    1,
    \(z) str_c(z[!is.na(z)], collapse = " | ")
  )

  header_rows <- which(
    map_lgl(
      seq_len(nrow(x)),
      \(row) any(
        str_to_upper(str_squish(unlist(x[row, ], use.names = FALSE))) ==
          "ANIMAL ID",
        na.rm = TRUE
      )
    )
  )

  # Determine the day attached to each repeated header. One D1 workbook block
# has a blank Day 30 label, so a second header following Day 1 is Day 30.
  block_days <- rep(NA_integer_, length(header_rows))

  for (i in seq_along(header_rows)) {
    first_context_row <- if (i == 1L) 1L else header_rows[i - 1L] + 1L

    day_labels <- str_extract_all(
      row_text[first_context_row:header_rows[i]],
      regex("\\bDAY\\s*(?:1|3\\d)\\b", ignore_case = TRUE)
    ) %>%
      unlist()

    block_days[i] <- if (length(day_labels) > 0L) {
      if_else(parse_number(last(day_labels)) == 1, 1L, 30L)
    } else if (i > 1L && block_days[i - 1L] == 1L) {
      30L
    } else {
```

```

    NA_integer_
  }
}

map2(header_rows, block_days, \(header_row, cue_day) {

  header <- x[header_row, ] %>%
    unlist(use.names = FALSE) %>%
    str_squish() %>%
    str_to_upper()

  id_col <- which(header == "ANIMAL ID")[1]
  condition_col <- which(header == "CONDITION")[1]
  infusion_col <- which(header == "INFUSIONS")[1]
  active_col <- which(header == "ACTIVE LEVER")[1]
  inactive_col <- which(header == "INACTIVE LEVER")[1]

  first_data_row <- header_row + 1L
  block_ids <- clean_subject(x[[id_col]][first_data_row:nrow(x)])
  first_blank <- which(is.na(block_ids))[1]

  last_data_row <- if (is.na(first_blank)) {
    nrow(x)
  } else {
    first_data_row + first_blank - 2L
  }

  out <- x[
    first_data_row:last_data_row,
    c(id_col, condition_col, infusion_col, active_col, inactive_col)
  ]

  names(out) <- c(
    "subject",
    "condition",
    "infusion",
    "active",
    "inactive"
  )

  needed <- missing_ids %>%
    filter(missing_cue, day == cue_day) %>%
    distinct(subject, reward, sex, day)

  out %>%
    mutate(
      subject = clean_subject(subject),
      condition = str_to_lower(str_squish(condition)),
      reward = "heroin",
      day = cue_day
    ) %>%
    filter(condition == "heroin") %>%
    inner_join(

```

```

      needed,
      by = c("subject", "reward", "day")
    ) %>%
    pivot_longer(
      c(infusion, active, inactive),
      names_to = "behavior",
      values_to = "value"
    ) %>%
    transmute(
      subject,
      reward,
      sex,
      phase = "cue_reactivity",
      day,
      behavior,
      value = parse_double(value)
    )
  }) %>%
  list_rbind()
}

```

Extract only previously missing records and append non-overlapping keys

```

trap_raw <- trap_files %>%
  map(\(file) {
    cohort_sheets <- excel_sheets(file) %>%
      keep(
        ~ str_detect(
          .x,
          regex(
            "(D1 TRAP|D2 TRAP|MIXED D1 D2 TRAP)",
            ignore_case = TRUE
          )
        )
      )
    )

    bind_rows(
      map(cohort_sheets, \(sheet) read_trap_ivsa(file, sheet)) %>%
        list_rbind(),
      read_trap_cue(file)
    )
  }) %>%
  list_rbind()

```

```

## New names:
## New names:
## New names:
## New names:
## New names:
## New names:
## New names:
## New names:
## New names:
## New names:
## New names:
## New names:
## * ' -> '...1'
## * ' -> '...2'
## * ' -> '...3'
## * ' -> '...4'
## * ' -> '...5'
## * ' -> '...6'
## * ' -> '...7'
## * ' -> '...8'
## * ' -> '...9'
## * ' -> '...10'
## * ' -> '...11'
## * ' -> '...12'
## * ' -> '...13'
## * ' -> '...14'

```

```

behavior_key <- c(
  "subject", "reward", "sex", "phase", "day", "behavior"
)

if (
  trap_raw %>%
    count(across(all_of(behavior_key))) %>%
    filter(n > 1L) %>%
    nrow() > 0L
) {
  stop("The TRAP import produced duplicate subject/day/behavior keys.")
}

behavior_raw <- bind_rows(
  behavior_raw,
  trap_raw %>%
    anti_join(
      behavior_raw %>% distinct(across(all_of(behavior_key))),
      by = behavior_key
    )
) %>%
  arrange(reward, sex, subject, phase, day, behavior)

```

Recalculate missingness from the updated behavior_raw and export the audit

```
missing_audit <- animal_ids %>%
  transmute(
    subject = clean_subject(subject),
    reward = str_to_lower(str_squish(reward)),
    sex = str_to_lower(str_squish(sex)),
    day = if_else(parse_number(as.character(day)) == 1, 1L, 30L)
  ) %>%
  left_join(
    behavior_raw %>%
      filter(
        phase == "ivsa",
        behavior %in% c("infusion", "active", "inactive")
      ) %>%
      group_by(subject, reward, sex, behavior) %>%
      summarise(
        n_days = n_distinct(day[!is.na(value)]),
        .groups = "drop"
      ) %>%
      pivot_wider(
        names_from = behavior,
        values_from = n_days,
        names_prefix = "n_",
        values_fill = 0L
      ),
    by = c("subject", "reward", "sex")
  ) %>%
  left_join(
    behavior_raw %>%
      filter(
        phase == "cue_reactivity",
        behavior %in% c("infusion", "active", "inactive")
      ) %>%
      group_by(subject, reward, sex, day) %>%
      summarise(
        n_cue_behaviors = n_distinct(behavior[!is.na(value)]),
        .groups = "drop"
      ),
    by = c("subject", "reward", "sex", "day")
  ) %>%
  mutate(
    across(
      c(n_infusion, n_active, n_inactive, n_cue_behaviors),
      ~ replace_na(.x, 0L)
    ),
  ),
```

```

missing_infusion = n_infusion < 10L,
missing_active = n_active < 10L,
missing_inactive = n_inactive < 10L,
missing_cue = n_cue_behaviors < 3L,
missing_data_types = pmap_chr(
  list(
    missing_infusion,
    missing_active,
    missing_inactive,
    missing_cue
  ),
  \(inf, act, inact, cue) {
    str_c(
      c(
        if (inf) "ivsa infusion",
        if (act) "ivsa active",
        if (inact) "ivsa inactive",
        if (cue) "cue reactivity"
      ),
      collapse = "; "
    )
  }
)

final_missing_by_row <- missing_audit %>%
  filter(
    missing_infusion |
    missing_active |
    missing_inactive |
    missing_cue
  ) %>%
  mutate(day = str_c("day ", day)) %>%
  select(
    subject, reward, sex, day,
    missing_infusion, missing_active,
    missing_inactive, missing_cue,
    missing_data_types
  )

final_missing_unique_animals <- missing_audit %>%
  group_by(subject, reward, sex) %>%
  summarise(
    days_in_animal_ids = str_c(
      "day ",
      sort(unique(day)),
      collapse = ", "
    ),
    missing_infusion = any(missing_infusion),
    missing_active = any(missing_active),
    missing_inactive = any(missing_inactive),
    missing_cue_day1 = any(day == 1L & missing_cue),
    missing_cue_day30 = any(day == 30L & missing_cue),

```

```

missing_any = any(
  missing_infusion |
  missing_active |
  missing_inactive |
  missing_cue
),
.groups = "drop"
) %>%
filter(missing_any)

write_xlsx(
  list(
    final_missing_by_row = final_missing_by_row,
    final_missing_unique = final_missing_unique_animals
  ),
  file.path(data_dir, "final_missing_animal_ids.xlsx")
)

```

6. plot drug by ivsa behavior

```

behavior_raw %>%
  filter(
    phase == "ivsa",
    behavior %in% c("active", "inactive", "infusion")
  ) %>%
  mutate(
    reward = factor(reward, levels = c("sucrose", "heroin")),
    behavior = factor(
      behavior,
      levels = c("active", "inactive", "infusion"),
      labels = c("Active Lever", "Inactive Lever", "Infusions")
    ),
    day = factor(day, levels = 1:10)
  ) %>%
  ggplot(
    aes(
      x = day,
      y = value,
      fill = reward
    )
  ) +

  # Group means
  stat_summary(
    fun = mean,

```

```

    geom = "col",
    position = position_dodge(width = 0.8),
    width = 0.7,
    alpha = 0.4
  ) +

  # Individual animals
  geom_point(
    aes(color = reward),
    position = position_jitterdodge(
      jitter.width = 0.15,
      dodge.width = 0.8
    ),
    size = 1.7,
    alpha = 0.15
  ) +

  facet_wrap(
    ~behavior,
    ncol = 1,
    scales = "free_y"
  ) +

  labs(
    x = "IVSA Day",
    y = "Response",
    fill = "Reward",
    color = "Reward"
  ) +

  theme_classic(base_size = 14) +
  theme(
    strip.background = element_blank(),
    strip.text = element_text(face = "bold"),
    legend.position = "top"
  ) -> ivsa_plot

ggsave("ivsa_data.png", plot = ivsa_plot)

```

Saving 6.5 x 4.5 in image

```
print(ivsa_plot)
```



7. summarizing ivsa data and merging w inc_sneek

```
inc_ivsa <- inc_sneek %>%
  left_join(

    behavior_raw %>%
      filter(
        phase == "ivsa",
        behavior %in% c("infusion", "active", "inactive")
      ) %>%

      mutate(
        period = if_else(day <= 5, "early", "late"),

        behavior = recode(
          behavior,
          infusion = "inf",
```

```

      active   = "act",
      inactive = "inact"
    )
  ) %>%

  group_by(subject, behavior, period) %>%

  summarise(
    avg = mean(value, na.rm = TRUE),
    sum = sum(value, na.rm = TRUE),
    .groups = "drop"
  ) %>%

  pivot_wider(
    names_from = c(behavior, period),
    values_from = c(avg, sum),
    names_glue = "{behavior}_{.value}_{period}"
  ),

  by = "subject"
)

inc_ivsa <- inc_ivsa %>%
  mutate(
    inf_sum = inf_sum_early + inf_sum_late,
    act_sum = act_sum_early + act_sum_late,
    inact_sum = inact_sum_early + inact_sum_late)

# collapse across the 6 timepoints
inc_model <- inc_ivsa %>%
  group_by(subject, drug, sex, session) %>%
  summarise(
    lever = sum(lever, na.rm = TRUE),
    beam = sum(beam, na.rm = TRUE),
    sneek = sum(sneek, na.rm = TRUE),
    groom = sum(groom, na.rm = TRUE),
    shake = sum(shake, na.rm = TRUE),

    inf_avg_early = first(inf_avg_early),
    inf_avg_late = first(inf_avg_late),
    act_avg_early = first(act_avg_early),
    act_avg_late = first(act_avg_late),
    inact_avg_early = first(inact_avg_early),
    inact_avg_late = first(inact_avg_late),

    inf_sum_early = first(inf_sum_early),
    inf_sum_late = first(inf_sum_late),
    act_sum_early = first(act_sum_early),
    act_sum_late = first(act_sum_late),
    inact_sum_early = first(inact_sum_early),
    inact_sum_late = first(inact_sum_late),

    inf_sum = inf_sum_early + inf_sum_late,

```

```

act_sum = act_sum_early + act_sum_late,
inact_sum = inact_sum_early + inact_sum_late,

.groups = "drop"
)

```

```

behavior_raw %>%
  filter(behavior == "active" & phase == "ivsa") -> active_levers

mod_act <- lmer(value ~ day * reward + (1|subject), data = active_levers)
summary(mod_act)

```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ day * reward + (1 | subject)
## Data: active_levers
##
## REML criterion at convergence: 11512.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.4669 -0.5277 -0.0475  0.5055  7.7576
##
## Random effects:
##  Groups   Name                Variance Std.Dev.
## subject (Intercept) 2104         45.87
## Residual                2498         49.98
## Number of obs: 1060, groups:  subject, 101
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)    54.471     9.955 189.367   5.472 1.40e-07 ***
## day              7.994     0.958 957.165   8.344 2.48e-16 ***
## rewardsucrose   38.934    12.088 186.762   3.221 0.00151 **
## day:rewardsucrose -2.085     1.154 957.165  -1.806 0.07118 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) day    rwrds
## day          -0.529
## rewardsucrs -0.824  0.436
## dy:rwrdsrscs  0.439 -0.830 -0.525

```

```

plot_model(mod_act, type = "pred")

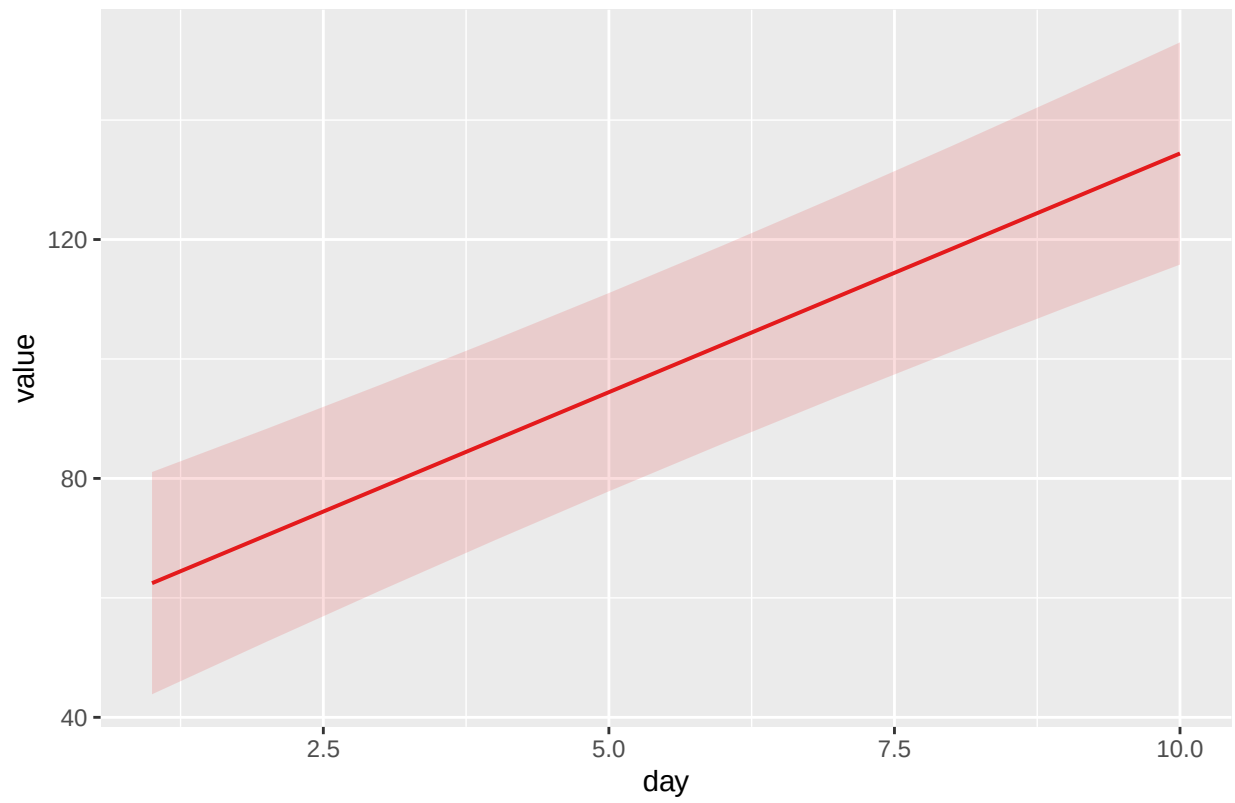
```

```

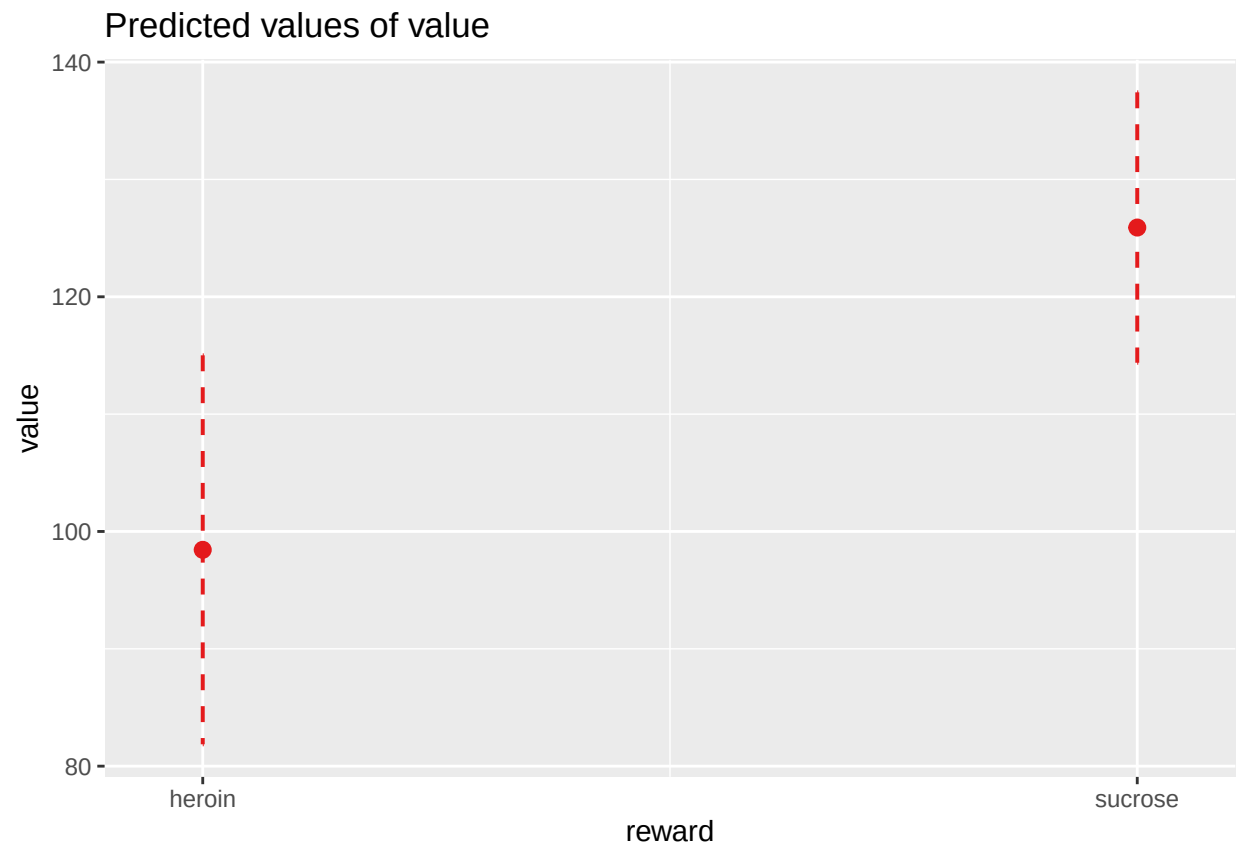
## $day

```

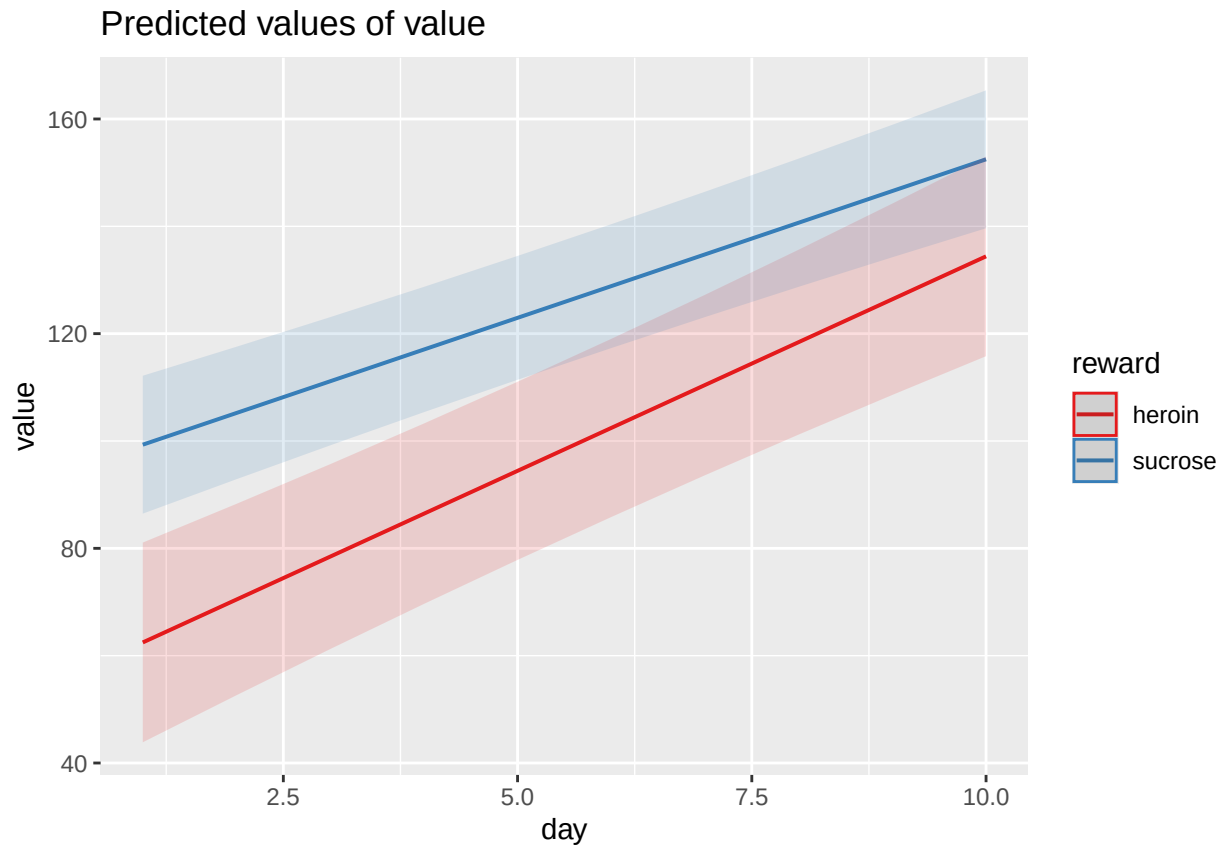
Predicted values of value



```
##  
## $reward
```



```
plot_model(mod_act, type = "int")
```



```
behavior_raw %>%
  filter(behavior == "inactive" & phase == "ivsa") -> inactive_levers

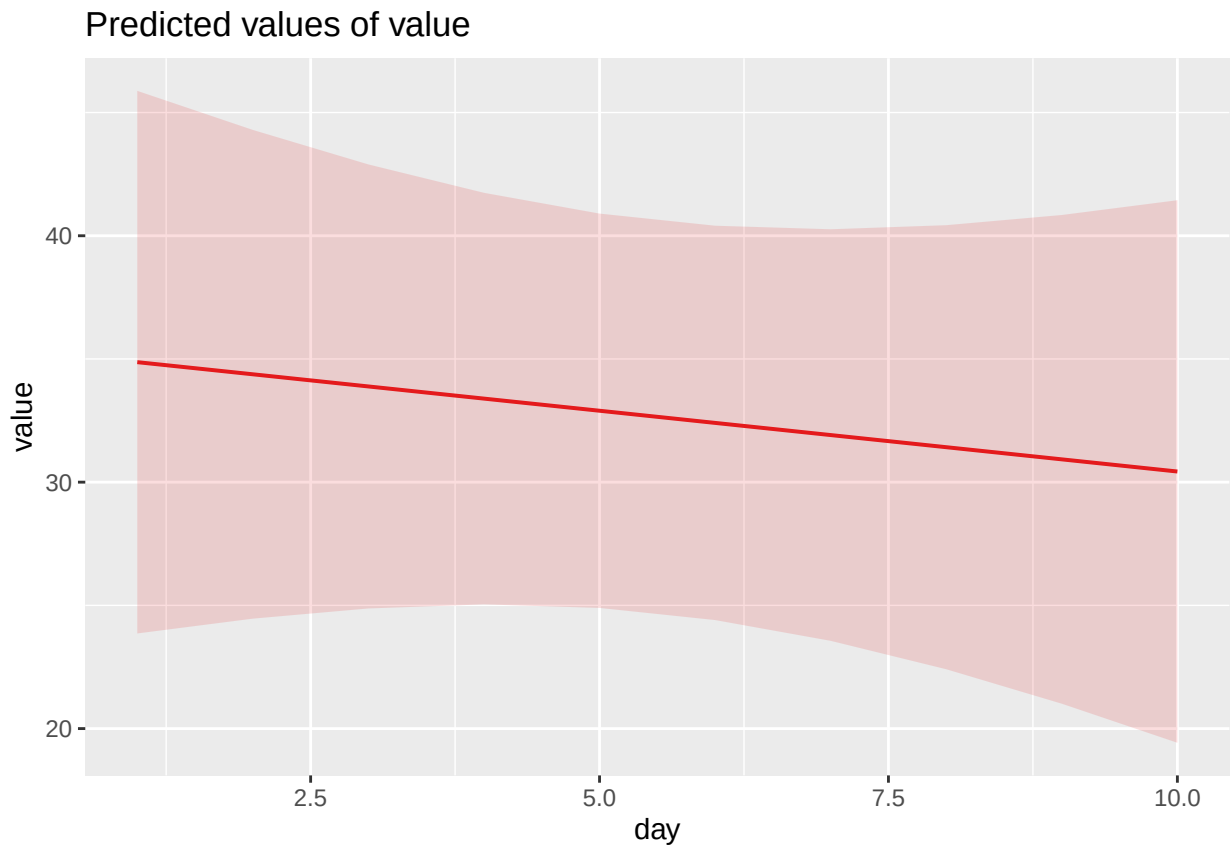
mod_inact <- lmer(value ~ day * reward + (1|subject), data = inactive_levers)
summary(mod_inact)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ day * reward + (1 | subject)
## Data: inactive_levers
##
## REML criterion at convergence: 11163.9
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -2.3325 -0.2305 -0.0833 0.0681 24.1456
##
## Random effects:
## Groups Name Variance Std.Dev.
## subject (Intercept) 340.5 18.45
## Residual 2021.7 44.96
## Number of obs: 1060, groups: subject, 101
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
```

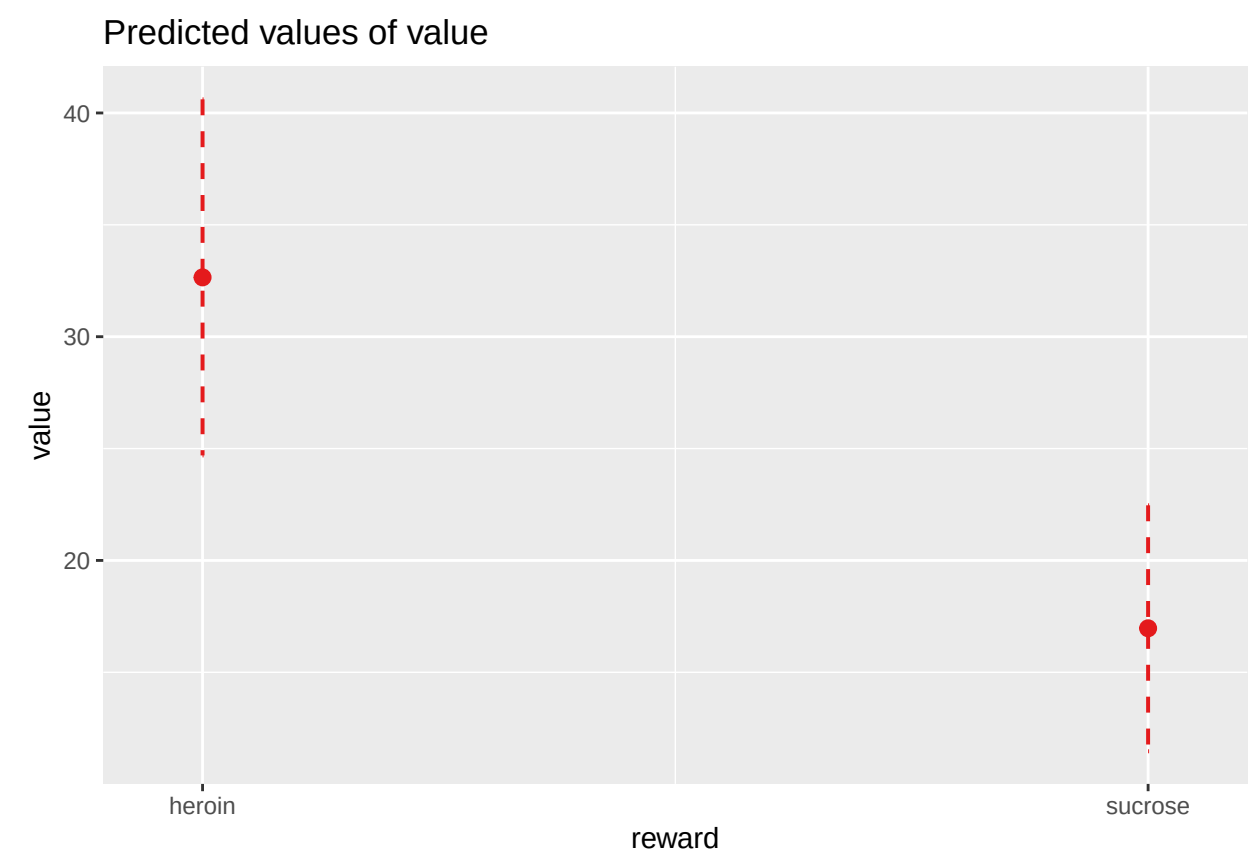
```
## (Intercept)      35.3636      6.2377 465.5071   5.669 2.52e-08 ***
## day              -0.4931      0.8617 954.7150  -0.572   0.567
## rewardsucrose    -6.8298      7.5441 454.5218  -0.905   0.366
## day:rewardsucrose -1.6101      1.0384 954.7150  -1.551   0.121
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) day    rwrds
## day      -0.760
## rewardsucrs -0.827  0.628
## dy:rwrdsrs  0.631 -0.830 -0.757
```

```
plot_model(mod_inact, type = "pred")
```

```
## $day
```

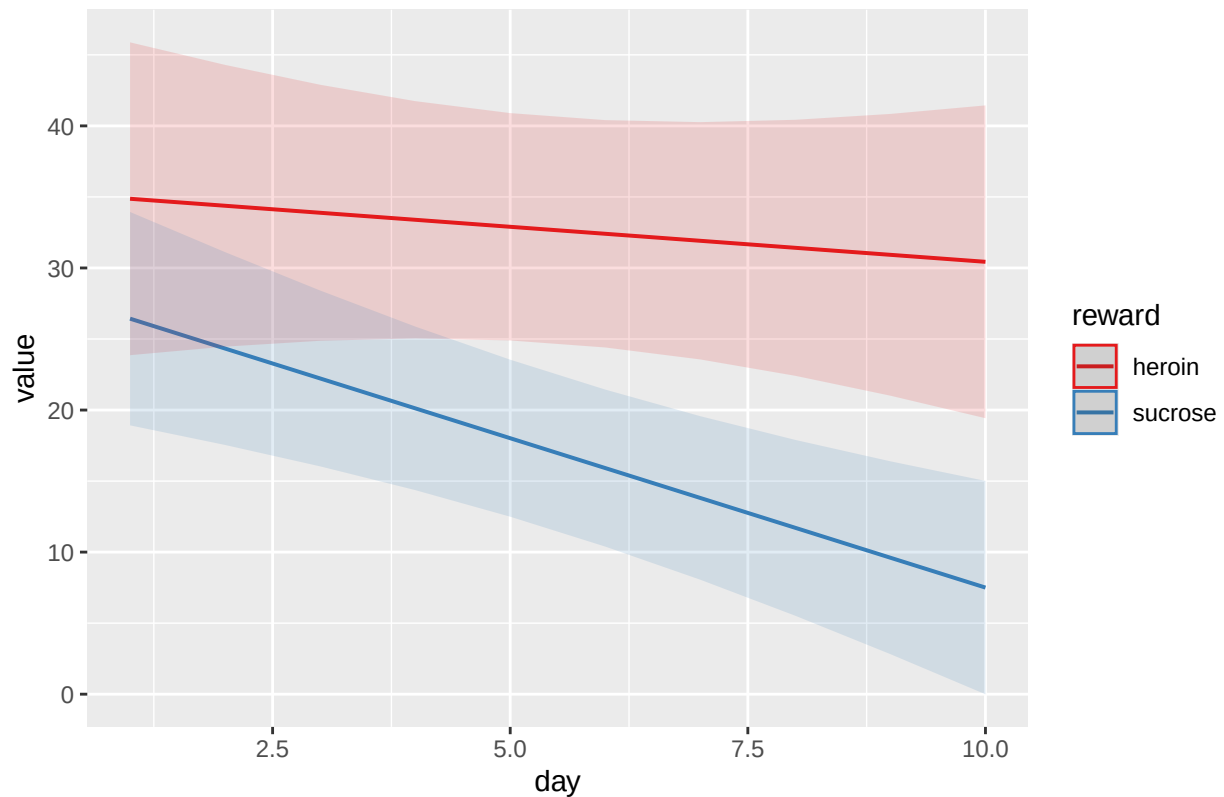


```
##
## $reward
```



```
plot_model(mod_inact, type = "int")
```

Predicted values of value



```
behavior_raw %>%
  filter(behavior == "infusion" & phase == "ivsa") -> infusions

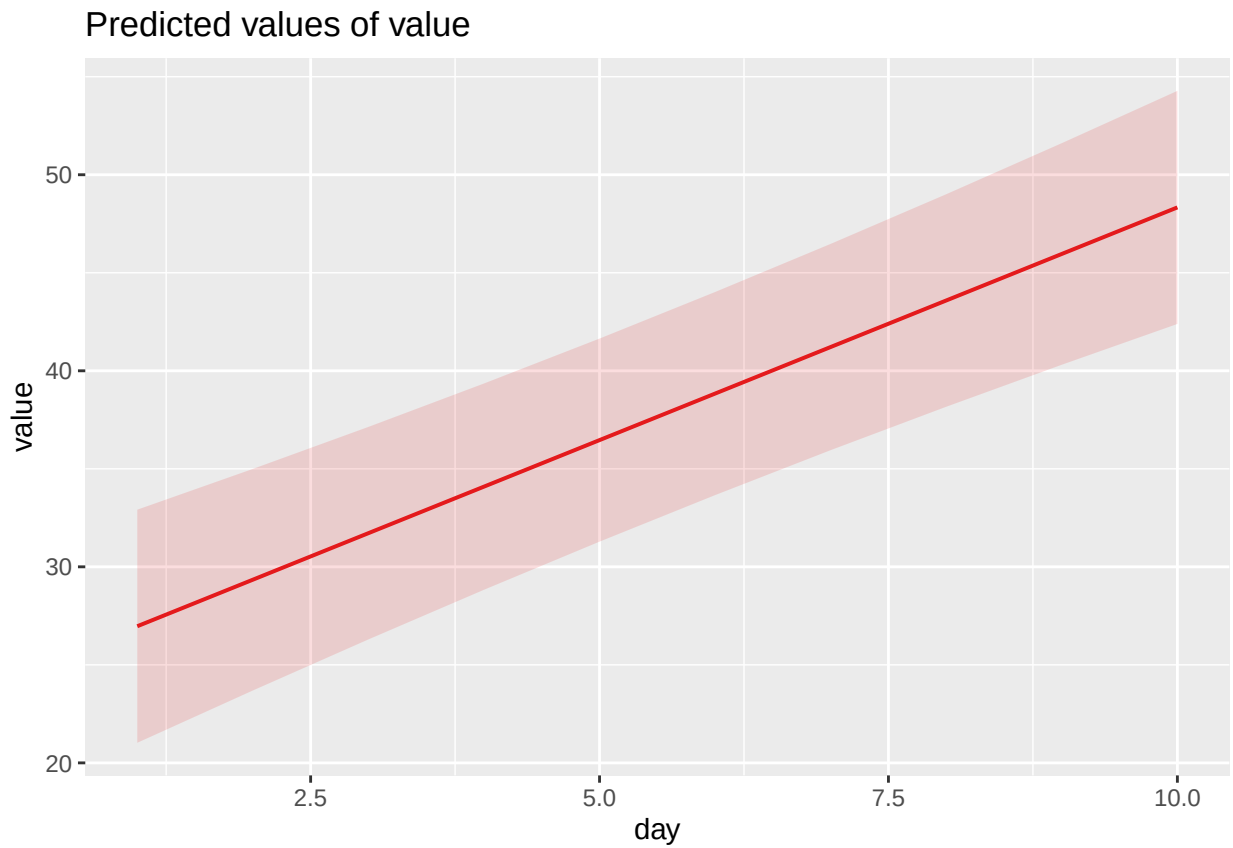
mod_inf <- lmer(value ~ day * reward + (1|subject), data = infusions)
summary(mod_inf)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ day * reward + (1 | subject)
## Data: infusions
##
## REML criterion at convergence: 9257.1
##
## Scaled residuals:
## Min      1Q  Median      3Q      Max
## -3.6354 -0.5269  0.0670  0.6087  3.1790
##
## Random effects:
## Groups Name Variance Std.Dev.
## subject (Intercept) 198.9 14.10
## Residual 301.2 17.35
## Number of obs: 1060, groups: subject, 101
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
```

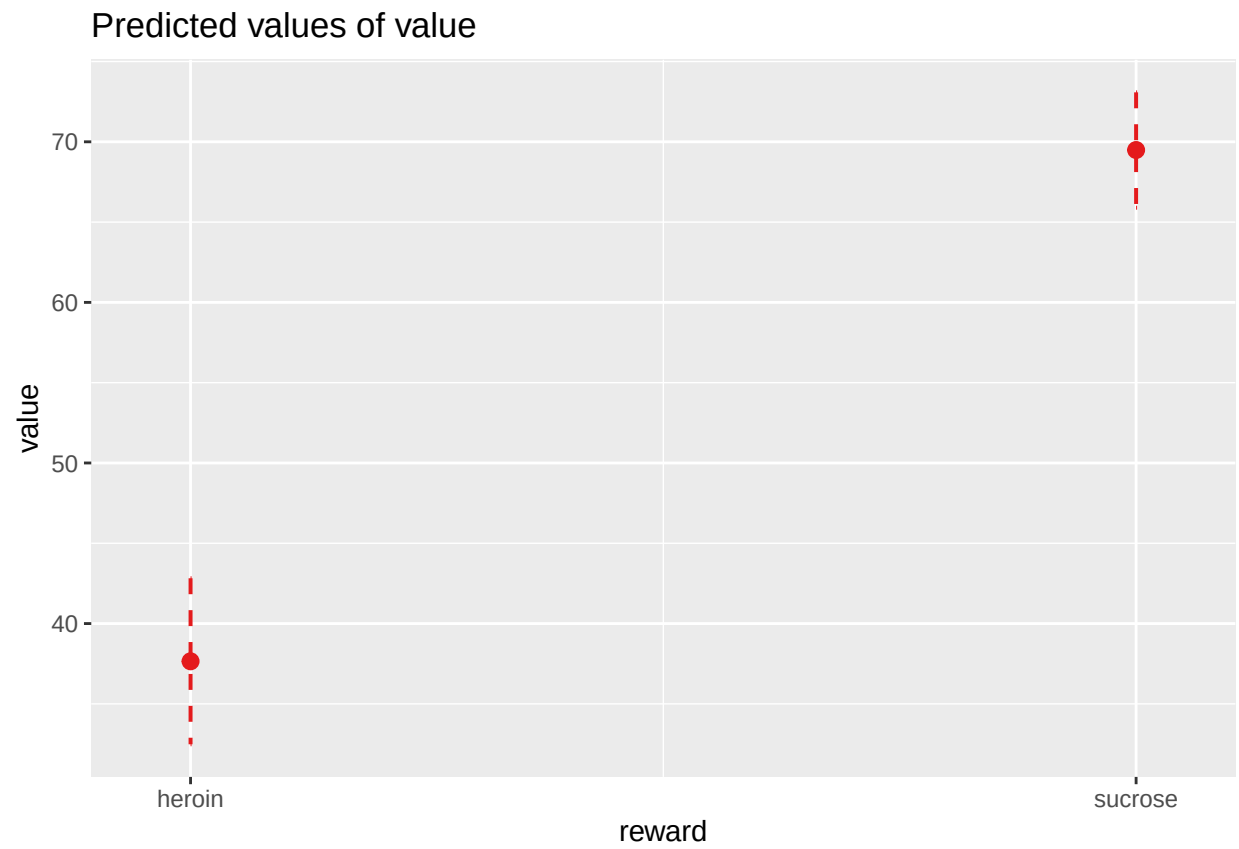
```
## (Intercept)      24.5960      3.2074 216.8571   7.669 5.79e-13 ***
## day              2.3737      0.3326 958.6099   7.137 1.89e-12 ***
## rewardsucrose    26.4462      3.8924 213.4514   6.794 1.07e-10 ***
## day:rewardsucrose 0.9801      0.4008 958.6099   2.445 0.0146 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) day    rwrds
## day      -0.570
## rewardsucrs -0.824  0.470
## dy:rwrdsrscs 0.473 -0.830 -0.566
```

```
plot_model(mod_inf, type = "pred")
```

```
## $day
```

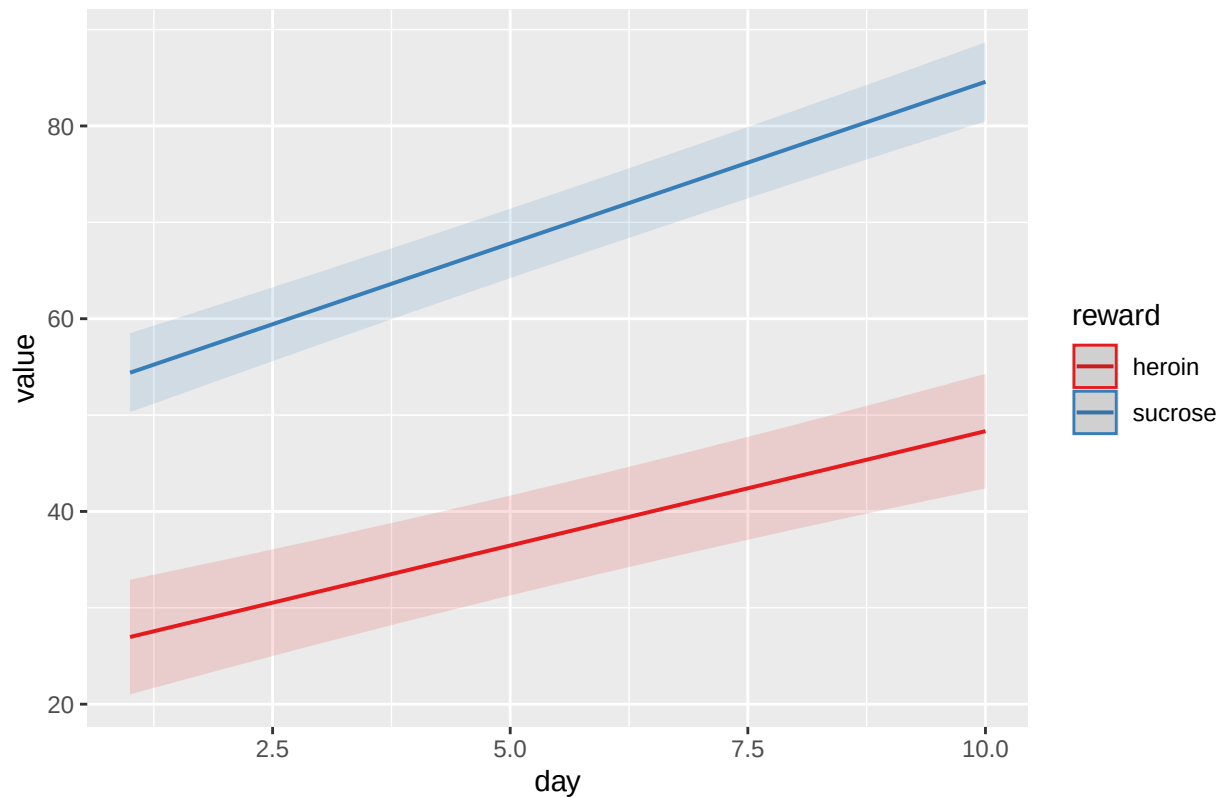


```
##
## $reward
```



```
plot_model(mod_inf, type = "int")
```

Predicted values of value



```
mod_ivsa <- lmer(lever ~ drug*session + (1|subject),data = inc_ivsa)
summary(mod_ivsa)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: lever ~ drug * session + (1 | subject)
## Data: inc_ivsa
##
## REML criterion at convergence: 4326.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.62343 -0.68386 -0.05141  0.59790  3.16100
##
## Random effects:
## Groups Name Variance Std.Dev.
## subject (Intercept) 0.5879  0.7668
## Residual          4.3320  2.0813
## Number of obs: 984, groups: subject, 126
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -3.0691     0.7776 575.0362  -3.947 8.89e-05 ***
## drug           3.2422     0.5117 389.0165   6.336 6.52e-10 ***
## session       2.3855     0.4708 897.0468   5.067 4.90e-07 ***
```

```
## drug:session -0.8409      0.3083 554.9218 -2.728 0.00658 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) drug  sessin
## drug      -0.945
## session   -0.903  0.860
## drug:session 0.864 -0.921 -0.946
```

```
anova(mod_ivsa)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## drug      173.919 173.919      1 389.02 40.1478 6.518e-10 ***
## session   111.233 111.233      1 897.05 25.6773 4.900e-07 ***
## drug:session 32.227 32.227      1 554.92 7.4395 0.006583 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mod_ivsa_beh <- lmer(lever ~ session*drug + inf_sum + (1|subject),data = inc_ivsa)
summary(mod_ivsa_beh)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: lever ~ session * drug + inf_sum + (1 | subject)
## Data: inc_ivsa
##
## REML criterion at convergence: 3442.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.49510 -0.66306 -0.03688  0.62961  3.02913
##
## Random effects:
## Groups Name Variance Std.Dev.
## subject (Intercept) 0.212 0.4604
## Residual 4.561 2.1357
## Number of obs: 780, groups: subject, 101
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -3.538e+00  8.543e-01 5.070e+02 -4.141 4.05e-05 ***
## session      2.454e+00  5.325e-01 7.023e+02 4.608 4.82e-06 ***
## drug         3.009e+00  5.743e-01 2.888e+02 5.240 3.10e-07 ***
## inf_sum      2.093e-03  6.312e-04 9.679e+01 3.316 0.00129 **
## session:drug -1.025e+00  3.310e-01 4.843e+02 -3.098 0.00206 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) sessin drug  inf_sm
## session   -0.931
```

```
## drug      -0.871  0.813
## inf_sum   -0.055  0.036 -0.343
## session:drg 0.896 -0.951 -0.864 -0.049
```